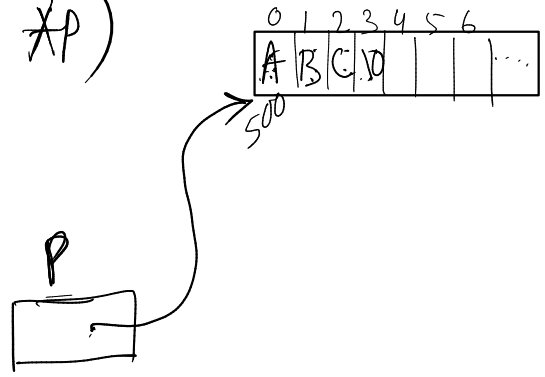
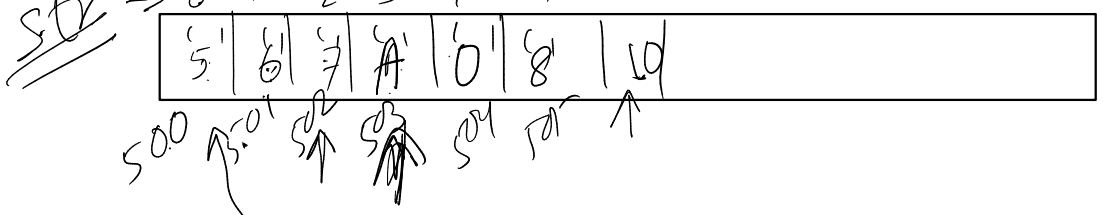
`unsigned long int Vstrlen(char *);` Declaration/Prototype

create ~~user-defined~~ function Vstrlen, to perform the same work as strlen library function

```
unsigned long int Vstrlen(char *p)
{
un    int i;
for(i=0; p[i] ; i++);
    return i;
}
```





int i

i
0
1
2
3
⋮

char *p

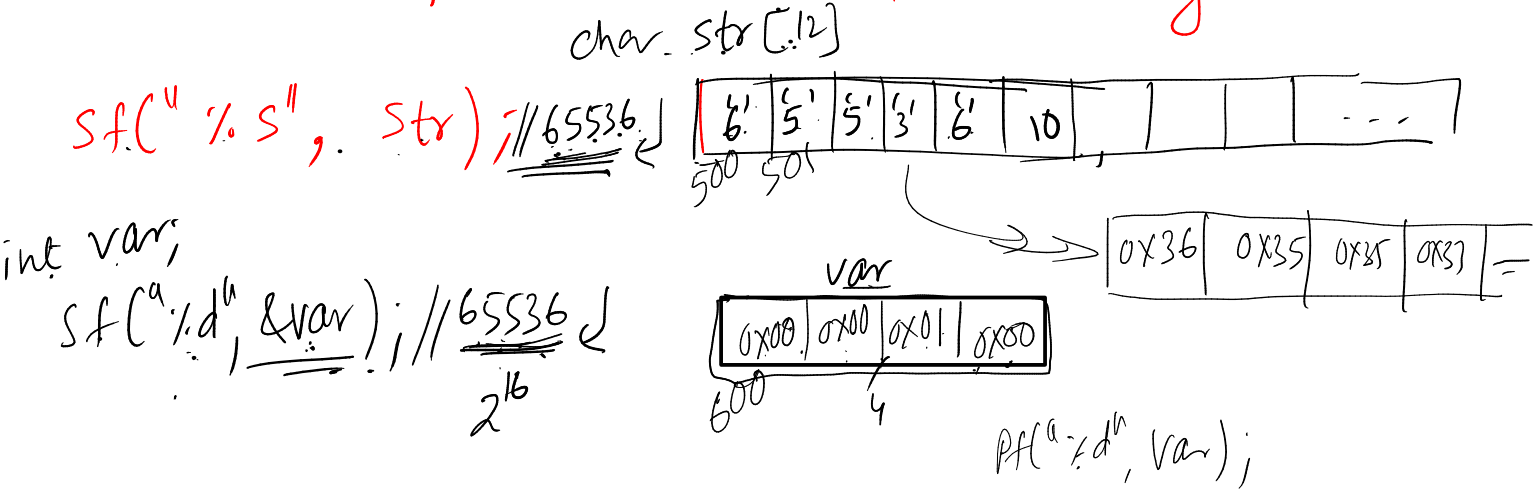
p
~~500~~
~~501~~
502

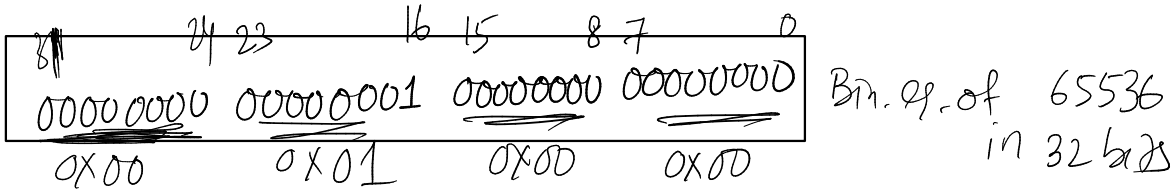



```
while(*p != '\0')
{
  ===
  ===
  p++;
}
```

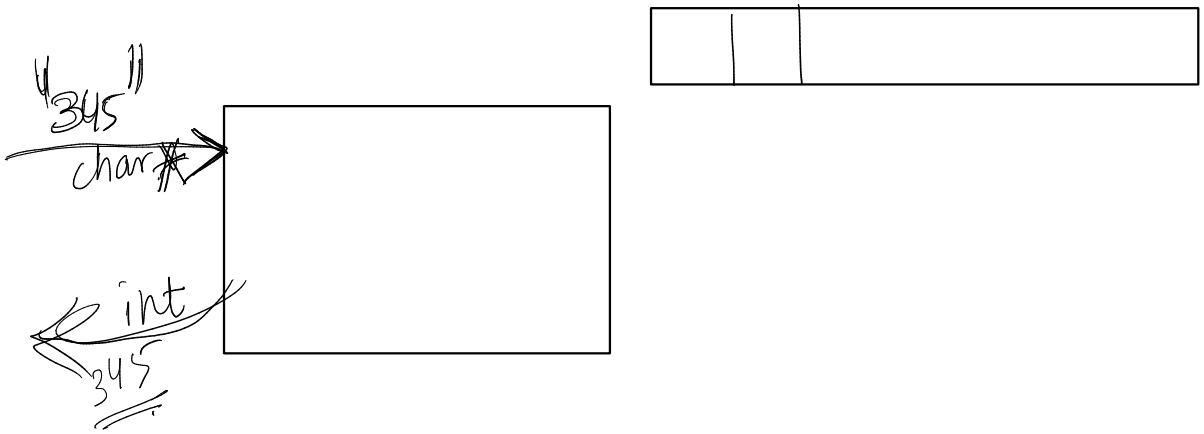
printf prints ONLY strings

scanf receives ONLY strings





Implement atoi



str

0	1	2	3	4		
'3'	'8'	'5'	10			

500 501

n = n*10 + (str[i] - 48) = 3

n = 3*10 + str[1] - 48 = 38

n = 38*10 + str[2] - 48 = 385

n: 385

Array / string Assignment / copy

```
char s1[20] = "HELLO", s2[20];
```

// s2 = s1; ✗

```
for(i=0; s1[i]; i++)
    s2[i] = s1[i];
```
✓

```
s2[i] = '\0';
```

s2[0] = s1[0]

s2[1] = s1[1];

⋮

s2	0	1	2	3	4	i
	H	E	L	L	0	\0

```
i=0;
do {
    s2[i] = s1[i];
} while(s2[i++]);
```
✓

s1

0	1	2	3	4	5	6
A	B	C	\0			

s2

0	1	2	3	4	5	6	7
A	B	C	\0				

i=4

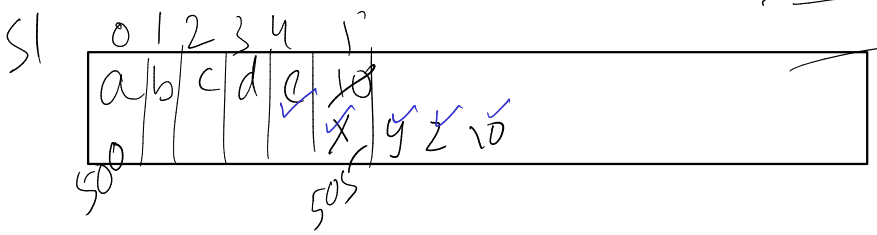
✓ char *strcpy(char *dest, const char * src);

strcpy
strncpy
strcat
memcpy
memmove

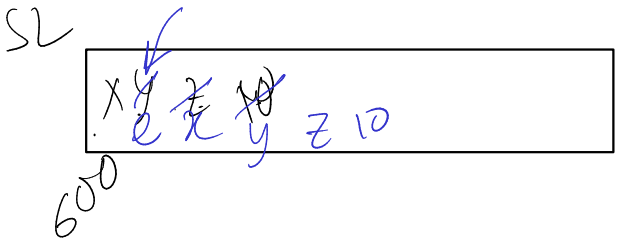
strcpy("abc", "xyz"); //Runtime Err seg fault

s2 = s1

strcpy(s2, s1);



All chars (including \0) is copied to s2.



strcpy(s1 + len(s1), s2);
505

strcpy(s2+1, s1+4);
601 504
src

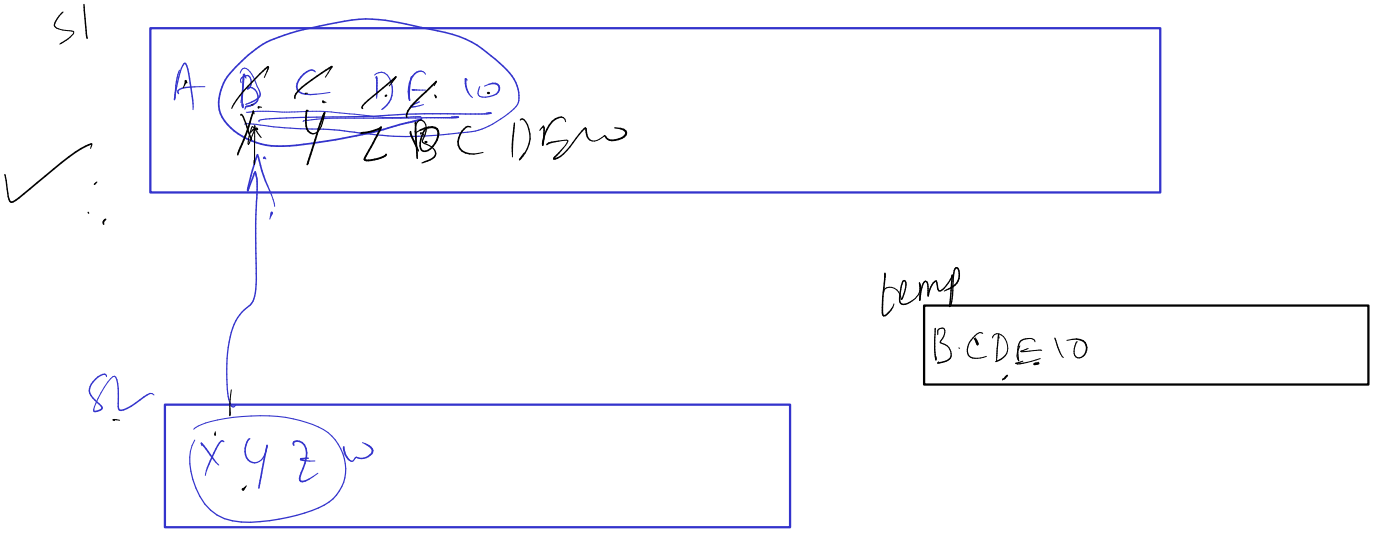
Task

W.A.P to insert string s2 at the specified posn in s1.

s1 ⇒ "abcdefg" ⇒ "axyzbcdefg"

s2 ⇒ "xyz"

posn ⇒ 1



I

↳ Step1 => cpy the portion of $1 to a temp Array

↳ Step2 => cpy S2 to the posn in S1

↳ Step3 : cpy temp at the last of S1

II : without using temp Array